

Chengle Fan

Undergraduate Student, Optoelectronic Information Science and Engineering
Southern University of Science and Technology, Shenzhen, China
standingfloating@gmail.com | +86 136 4265 8984

EDUCATION

Southern University of Science and Technology

Shenzhen, China

B.Eng. in Optoelectronic Information Science and Engineering

Expected June 2027

- GPA: 3.85/4.00; Weighted Average: 92.45/100; Major Rank: 7/46.
- Selected coursework: Fundamentals of Optics, Engineering Electromagnetics, Solid-State Electronics, Introduction to Semiconductor Optics, Liquid Crystal Optoelectronics, Optical System Design Experiment.

RESEARCH INTERESTS

Artificial photonic structures and nanophotonic systems, including photonic crystals, metamaterials, and metasurfaces, for controlling light propagation and engineering light-matter interactions. Specific interests include topological and time-varying phenomena, nonreciprocity, and reconfigurable photonic functionalities.

MANUSCRIPT IN PREPARATION

Chengle Fan, Xiang Xi, and Zhen Gao, “Observation of pseudospin-locked chiral edge states in a gyromagnetic photonic crystal,” *Manuscript in preparation*.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Feb. 2025–Present

Topological Physics Research Group, Department of EEE, SUSTech

Supervisor: Prof. Zhen Gao

Pseudospin-Locked Chiral Edge States in Gyromagnetic Photonic Crystals

- Led full-wave modeling and sample design for a microwave gyromagnetic photonic crystal supporting pseudospin-locked chiral edge transport under broken time-reversal symmetry.
- Identified QSH-like, QAH-like, and trivial phases through bulk and projected band structures, p - d band inversion, pseudospin-resolved eigenfields, and spin Chern numbers.
- Designed measurement configurations and participated in VNA transmission and near-field mapping; measurements showed approximately 20 dB directional transmission contrast within the bandgap and pseudospin-selective excitation of the chiral edge mode.
- Designed a pseudospin multiplexer/demultiplexer and a low-crosstalk waveguide crossing based on the coexistence of multiple topological phases.

Gauge-Field-Induced Duality and Hofstadter Photonic Lattices

- Developed tight-binding and full-wave models of gauge-flux-engineered dual and self-dual photonic crystals, verifying duality-related spectral equivalence and self-duality-induced degeneracies.
- Engineered rotation-dependent effective couplings of higher-order photonic modes to implement spatially modulated hopping amplitudes and investigate Hofstadter-butterfly-like spectra.
- Built finite-array simulations and MATLAB workflows for spectral reconstruction and Fourier analysis, and explored how synthetic gauge flux, coupling modulation, and duality affect the resulting fractal band structure.

Cavity-Mediated Topological Transport in Metallic Dipole Arrays

- Conduct microwave transmission and near-field measurements and process S-parameter and field data for an ongoing collaboration between Prof. Gao’s and Prof. Mengyao Li’s groups on cavity-tunable topological transport in all-metal dipole arrays; reconstruct dispersions by FFT and compare them with simulated bands.

Visiting Research Student

School of Physical and Mathematical Sciences, NTU

Jan. 2026–Apr. 2026

Supervisor: Prof. Baile Zhang

Terahertz Chiral Valley Edge States in Magneto-Optic Photonic Crystals

- Designed and numerically validated a fabrication-compatible terahertz Chern–valley photonic-crystal interface supporting chiral, valley-selective edge modes.
- Built 2D and 3D COMSOL bulk-band and supercell models to identify overlapping topological gaps and interface-localized modes.
- Evaluated magnetic-bias configurations, fabrication tolerances, domain alignment, and finite air-layer effects, and re-optimized the structure to provide parameters for subsequent fabrication and experiments by NTU collaborators.

RESEARCH GRANT

SUSTech Undergraduate Research Training Program

Project Leader

Shenzhen, China

June 2026–June 2027

- Secured a competitive university research grant of CNY 10,000 for “Design of 3D Photonic Topological Insulators at Telecommunication Wavelengths.”

TECHNICAL SKILLS AND RESEARCH METHODS

- **Computational electromagnetics:** COMSOL Multiphysics and MATLAB for eigenmode and frequency-domain modeling of photonic crystals, metamaterials, and metasurfaces, including bulk/projected bands, eigenfield reconstruction, finite structures, and dispersion analysis.
- **Band topology and numerical modeling:** Tight-binding modeling; Berry phase and curvature, Chern number, and Wilson-loop calculations; synthetic gauge fields, wave-packet dynamics, and FFT-based dispersion reconstruction.
- **Time-varying and spatiotemporal photonics (ongoing research training):** Undertaking structured training in Prof. Gao’s group under the guidance of a postdoctoral researcher, focusing on MATLAB-based PWE/Floquet–Bloch band calculations, eigenfield reconstruction, and FDTD simulations for photonic time crystals and space–time-periodic media.
- **Experimental characterization:** VNA-based microwave transmission spectroscopy, near-field mapping, chiral-source excitation, S-parameter acquisition and analysis, and spatial-Fourier reconstruction of dispersions; experience with optical alignment, CCD imaging, polarization microscopy, liquid-crystal cell assembly, and basic electro-optic measurements.
- **Programming and software:** MATLAB for numerical eigensolvers, time-domain simulations, data processing, visualization, and COMSOL LiveLink automation; basic experience with C and Meep FDTD; familiarity with Zemax and TracePro.
- **Languages:** Mandarin Chinese (native); English (TOEFL iBT 104).

HONORS AND AWARDS

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| • SUSTech Outstanding Student Leader | 2025 |
| • SUSTech Outstanding Student | 2024, 2025 |
| • SUSTech Excellence Scholarship, Second Class / Third Class | 2024, 2025 |

LEADERSHIP AND ACTIVITIES

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| • Deputy Head, Activities Department, Student Union, Department of EEE, SUSTech | 2024–2025 |
| • Vice President, SUSTech Orienteering Club | 2024–2025 |